

Nested Decisions

Look at this code:

```
raining = input("Is it raining? yes/no: ")
if raining == "yes":
    windy = input("Is it windy? yes/no: ")
    if windy == "yes":
        print("Wear a coat.")
    else:
        print("Take an umbrella.")
else:
    print("No rain gear needed.")
```

Use the code above to help you. After completing each challenge below, run your program to test that it works correctly. Save each completed challenge program in a new file.

Programming Challenges

Challenge 1

When [raining] is "yes", ask whether the wind is strong or light and store the answer in [wind_strength]. Display "Wear a coat." for "strong" and "Take an umbrella." for "light". Keep "No rain gear needed." when [raining] is "no". Test "yes" and "strong"; display "Wear a coat." Test "no"; display "No rain gear needed."

Challenge 2

Change the decision for when [raining] is "no". Ask whether it is sunny and store the answer in [sunny]. Display "Take sunglasses." when [sunny] is "yes"; otherwise display "No weather gear needed." Keep the two rain outputs unchanged. Test "no" and "yes"; display "Take sunglasses." Test "no" and "no"; display "No weather gear needed."

Challenge 3

Ask whether the visitor has a library card and store the answer in [has_card]. Use a nested if statement. When [has_card] is "yes", ask whether a book is overdue and store the answer in [overdue]. Display "Return the



overdue book first." when [overdue] is "yes"; otherwise display "You may borrow a book." When [has_card] is "no", display "Apply for a library card." Test "yes" and "yes", "yes" and "no", then "no"; verify all three outputs.